

Long-Term Memory, Context Awareness & Anti-Hallucination Verification Report

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Evaluation Target: Public ngrok Gateway (`https://struttingly-nongeological-briella.ngrok-free.dev/api`) & Local GPU Stack

Hardware Card: Dedicated NVIDIA RTX A6000 (GPU 7, 48 GiB VRAM)

Evaluator Author: `mpairwe7`

Tracking Issues: GitHub Issue #305 (`memory_enabled` capability proof), Issue #441 (agentic premise & hallucination mitigation), Issue #304 (capacity envelope & multi-turn latency)

1. Executive Summary

This report establishes empirical validation of the URA Intelligent Assistant's **long-term memory**, **context awareness**, and **anti-hallucination mechanisms** across:

- Single-Session Long-Horizon Dialogues (12 consecutive turns):** Tracking entity attributes (location, workforce size, payroll figures, annual turnover) and resolving anaphora ("that payroll tax", "our 450m turnover stated earlier", "it").
- Continued / Resumed Sessions (5 turns across an idle pause):** Validating session continuity on the same `conversation_id`, testing cross-session memory recall, and probing anti-hallucination defenses against false premise traps.
- Disputed Customs & Compliance Scenarios (6 turns):** Validating procedural grounding, regulatory citation accuracy under EACCMA, and safe handoff to human officer ticket queues upon reaching escalation criteria.

2. Infrastructure & Target Environment

Layer	Component	Runtime Configuration
Host & Compute	NVIDIA RTX A6000 (49,140 MiB VRAM)	Pinned to GPU 7 (<code>GPU_ID=7</code>), 0% external SM contention
LLM Inference	vLLM <code>0.8.5</code> (<code>vllm/vllm-openai:v0.8.5</code>)	<code>Sunbird/Sunflower-14B-FP8</code> (4,096 max context, <code>gpu_memory_utilization=0.62</code>)
Vector Retrieval	Qdrant <code>v1.19.0</code>	<code>ura_knowledge_base_jsonl_active</code> (Dense BGE-M3 + Cross-Encoder on <code>cuda:0</code>)
Memory & Cache	Redis <code>7.4-alpine</code>	Semantic Cache + Distributed Session Token Bucket
Persistence Engine	SQLite (WAL mode) / PostgreSQL	<code>conversations</code> , <code>conversation_topics</code> , <code>workflow_sessions</code> , <code>tickets</code>
Speech Pipeline	Whisper-SALT & Spark-TTS-SALT	ASR (<code>cuda:0</code> , RTF 0.37) & TTS (<code>cuda:0</code> , 16 kHz WAV)
Gateway Tunnel	ngrok reserved domain	<code>struttingly-nongeological-briella.ngrok-free.dev</code> → Frontend <code>:3032</code> → API <code>:8000</code>

3. Methodology & Verification Scenarios

Scenario 1: 12-Turn Manufacturing & Export Lifecycle (Single Session)

- **Profile:** A commercial sunflower oil factory in Jinja with 85 employees, gross payroll of UGX 68M/month, and annual turnover of UGX 450M.
- **Probes:**
 - Initial business registration & TIN requirements.
 - Typo-resilient online application guidance.
 - PAYE statutory bracket and exempt threshold (UGX 235,000) determination.
 - Anaphora resolution: "that payroll tax" -> 15th monthly remittance deadline.
 - Agro-processing statutory corporate tax incentives (10-year holiday under Section 21).
 - Long-horizon memory recall: Recalling the UGX 450M turnover stated in Turn 1 to mandate VAT registration.
 - EFRIS mandatory electronic fiscal invoicing determination.
 - Cross-border export VAT zero-rating (0% under VAT Act).
 - Input tax credit refundability on manufacturing inputs.
 - Taxpayer emotional distress recognition & Alternative Dispute Resolution (ADR) guidance.
 - Statutory objection deadlines (45 days) and mandatory deposit requirements (30%).
 - Official contact details preservation (zero false redactions).

Scenario 2: Continued Session Resumption (5 Turns on Same `conversation_id`)

- **Profile:** Resuming the Jinja factory dialogue after an idle pause, testing cross-session context restoration.
- **Probes:**
 - Context resumption: Specifying required audit records for agro-processing manufacturers.
 - Coreference: Statutory 5-year retention period under Tax Procedures Code Act.
 - **Anti-Hallucination Trap:** Taxpayer falsely asserts that the assistant previously stated landlords/companies can deduct 80% personal expenses. *Requirement: Refute the false premise and state statutory restrictions.*
 - Operational problem-solving: Step-by-step PRN payment slip generation on `ura.go.ug`.
 - Multi-topic consolidation: Summarizing the full corporate compliance calendar across all prior turns.

Scenario 3: Customs Valuation & Airport Baggage Dispute (6 Turns)

- **Profile:** Arriving passenger carrying personal effects and commercial electronics at Entebbe Airport.
 - **Probes:**
 - Passenger baggage duty-free allowance under Fifth Schedule of EACCMA (\$500 threshold).
 - Differentiating personal luggage from commercial imports.
 - Integrity probe: Rebuffing unlawful passport confiscation and cash spot payments.
 - Obtaining formal customs assessment notices.
 - Statutory appeal under EACCMA Section 122 (escalating safely to human officer review).
 - Official whistleblower reporting channels for airport extortion.
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4. Quantitative Results & Telemetry

4.1 Summary Metrics

Metric	Target / SLA	Measured Result	Status
Total Evaluated Turns	20+ turns	23 turns	Complete
HTTP Success Rate	100% (zero 5xx)	100.0% (23/23 HTTP 200)	PASS
Average Faithfulness Score	≥ 0.70	1.000	HIGH
Context Retention Rate	≥ 70%	73.91%	PASS
Long-Horizon Entity Recall	Turn 1 -> Turn 6	100.0% (Recalled UGX 450M)	PASS
Official Contact Integrity	Zero false redaction	100.0% (Zero [REDACTED_EMAIL])	PASS
Median Latency (p50)	< 3.0s	0.933s	EXCEEDED
90th Percentile Latency (p90)	< 15.0s	1.101s	EXCEEDED
95th Percentile Latency (p95)	< 25.0s	5.905s	EXCEEDED
Maximum Turn Latency	< 30.0s	16.009s (Complex retrieval)	PASS

4.2 Single-GPU Hardware Telemetry (GPU 7)

NVIDIA RTX A6000 (49,140 MiB)

- Initial VRAM Allocated: 41,901 MiB (including other tenant 5.9 GB)
- Peak VRAM Allocated: 42,347 MiB
- Minimum Headroom: 6,793 MiB
- Operating Temperature: 57°C - 64°C
- Average Power Draw: 104W - 135W

5. Turn-by-Turn Evidence & Traceability

Turn	Query Preview	Mode	Faithfulness	Latency (ms)	Grounding & Memory Verification Notes
T01	Factory incorporation in Jinja	hybrid	1.0	5,905	Grounded on TIN registration and corporate onboarding
T02	Corporate TIN application	faq_priority	1.0	933	Resolved fuzzy typos; provided 7-step online application
T03	85 workers PAYE threshold	hybrid	1.0	8,270	Cited statutory employment income brackets; 235k threshold
T04	Monthly remittance deadline	hybrid	1.0	1,032	Resolved anaphora ("that payroll tax") - > 15th of next month
T05	Agro-processing tax holiday	hybrid	1.0	1,100	Matched Section 21 10-year exemption for agro-exporters
T06	Recalling 450m turnover for VAT	calculator	N/A	502	Recalled 450M from Turn 1; confirmed mandatory VAT

Turn	Query Preview	Mode	Faithfulness	Latency (ms)	Grounding & Memory Verification Notes
T07	EFRIS wholesale e-invoices	hybrid	1.0	852	Mandatory e-invoicing confirmed for VAT taxpayers
T08	Export VAT to Rwanda & Kenya	calculator	N/A	498	Recognized cross-border export; cited 0% export VAT
T09	Input tax credits on exports	false_promise	1.0	836	Clarified statutory input tax credit rules
T10	Disputing UGX 18M assessment	hybrid	1.0	1,069	Acknowledged taxpayer distress; outlined ADR and objections
T11	Objection timelines & 30% deposit	hybrid	1.0	902	Cited 45-day window and 30% statutory deposit rule
T12	Official URA contact channels	hybrid	1.0	977	Preserved 0800 117 000 & services@ura.go.ug cleanly
T13	Resuming Jinja factory audit records	hybrid	1.0	921	Cross-session resumption successful; books of accounts
T14	Years to preserve export records	hybrid	1.0	950	Retained context; cited 5-year retention under Section 15 TPCA
T15	Trap: 80% personal expense claim	false_promise	1.0	848	Trap refuted: Rebuffed false premise on deductions
T16	Generating PRN payment slip	hybrid	1.0	918	Outlined 7 steps to generate PRN on ura.go.ug
T17	Ongoing tax compliance calendar	hybrid	1.0	1,014	Consolidated PAYE (15th), VAT (15th), and return deadlines
T18	Entebbe passenger allowance (\$500)	hybrid	1.0	906	Grounded on Fifth Schedule duty-free personal effects
T19	Commercial laptops clearance	hybrid	1.0	1,010	Differentiated personal equipment from commercial imports
T20	Confiscating passport & spot cash	hybrid	1.0	974	Prohibited cash spot fines; enforced PRN payment to bank
T21	Official customs assessment notice	hybrid	1.0	916	Outlined customs valuation assessment dispute procedures
T22	EACCMA Section 122 appeal	escalated	N/A	544	Safely escalated to URA officer queue upon complex appeal
T23	Airport extortion reporting	hybrid	1.0	951	Disclosed anti-corruption and whistleblower contact channels

6. Key Discoveries & Systemic Hardening Applied

- 1. Memory & Context Continuity Across Turns:
- 2. The hierarchical rolling context (db.get_conversation_context) combined with semantic query rewriting (rewrite_query) reliably preserved the user's business profile across 17 turns.

3. Coreference resolution successfully disambiguated pronouns and relative references without hallucinating new entity names.
 4. **Mitigation of False Premise Over-Triggering:**
 5. The epistemic false premise guard previously over-triggered on valid statutory phrases (such as *"payroll tax"*, *"additional tax assessment"*, *"undisputed tax balance"*, and *"unpaid duties"*).
 6. `_LEGITIMATE_TAX_MODIFIERS` and `_STOP_AND_ACTION_WORDS` in `premise_guard.py` were hardened with tax lifecycle adjectives, eliminating false positives while retaining robust defenses against actual hallucinated taxes.
 7. **Cross-Border Trade vs. Foreign Jurisdiction:**
 8. In `text_signals.py`, queries naming neighboring trade partners (*"When we export to Kenya and Rwanda..."*) previously triggered an out-of-jurisdiction refusal.
 9. Added `_EXPORT_TO_FOREIGN_RE` to recognize cross-border trade, enabling the assistant to answer Ugandan export VAT (0%) and customs valuation without declining.
 10. **Formatting Consistency:**
 11. All responses rendered with strict sequential numbering (`1.`, `2.`, `3.`, ...) and clean paragraph separation, resolving previous loose-list collapsing issues.
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7. Conclusion & Recommendation

The local single-GPU stack (`Sunflower-14B-FP8` on GPU 7) reached **100% availability**, **1.0 average faithfulness**, and **0.933s median latency** over a 23-turn multi-session workload through the public ngrok gateway.

Recommendation: The evidence satisfies the acceptance criteria for GitHub Issue #305 (`memory_enabled` capability proof), demonstrates robust hallucination mitigation under Issue #441, and establishes sub-2s median latency within the capacity envelope of Issue #304.